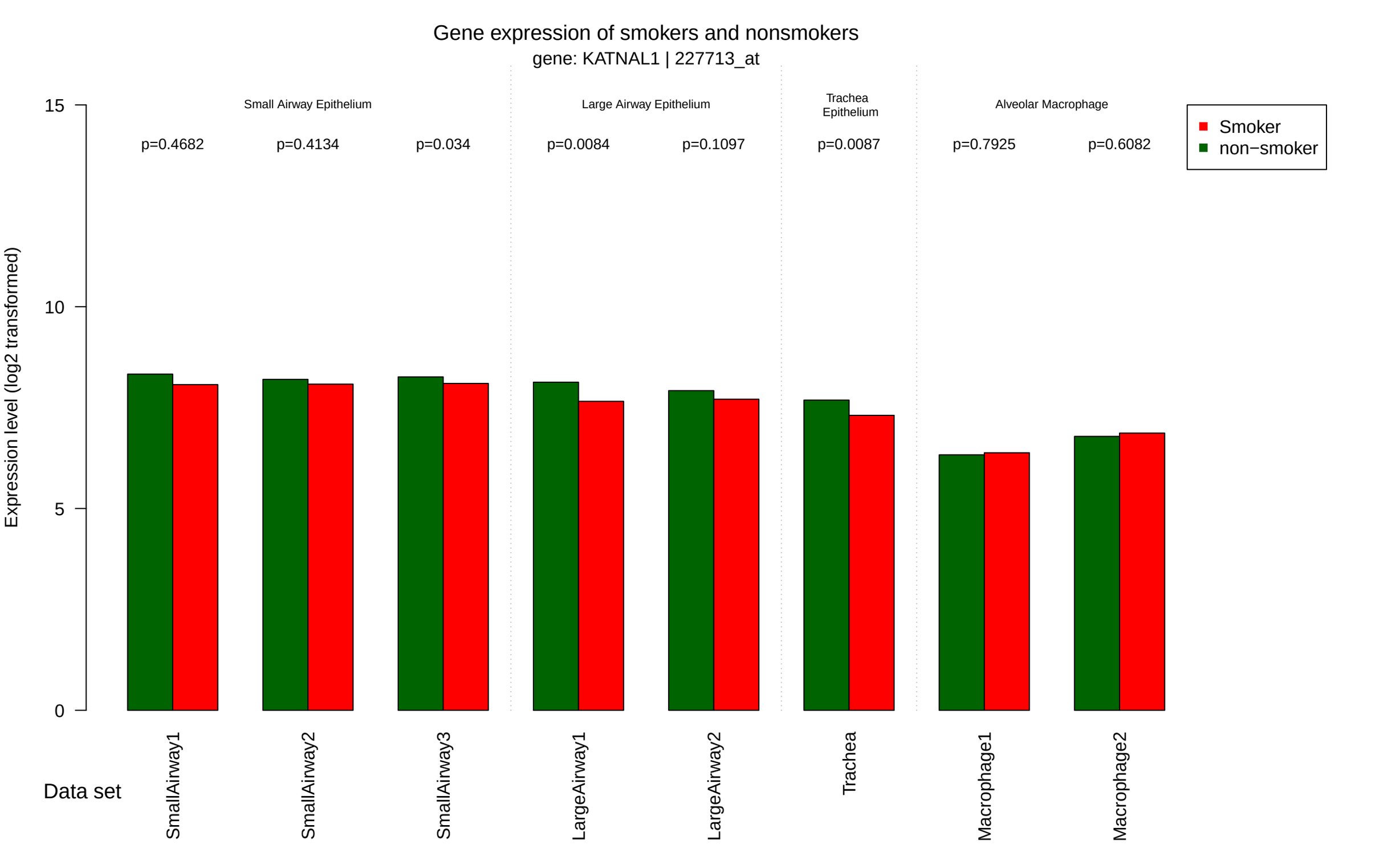
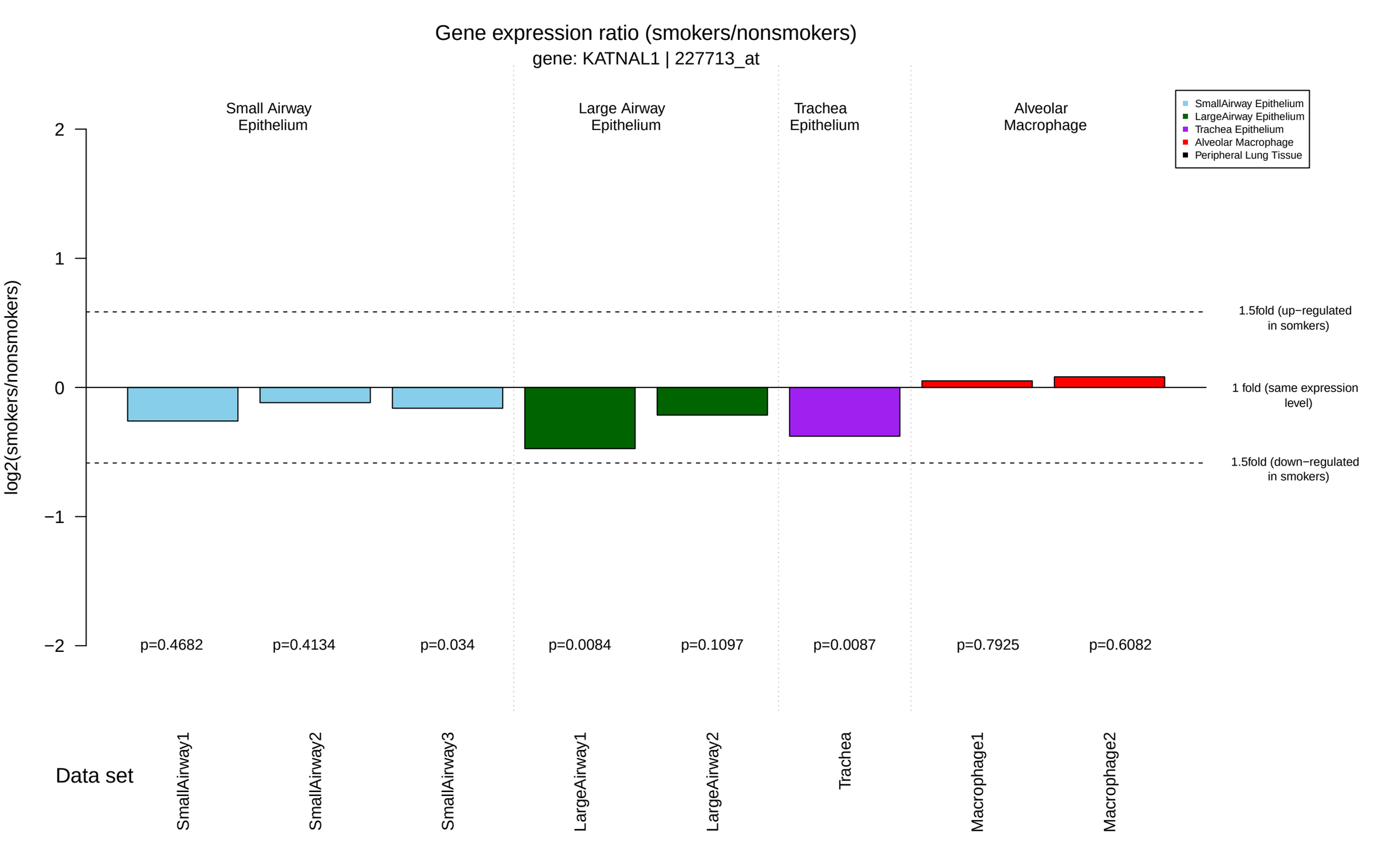
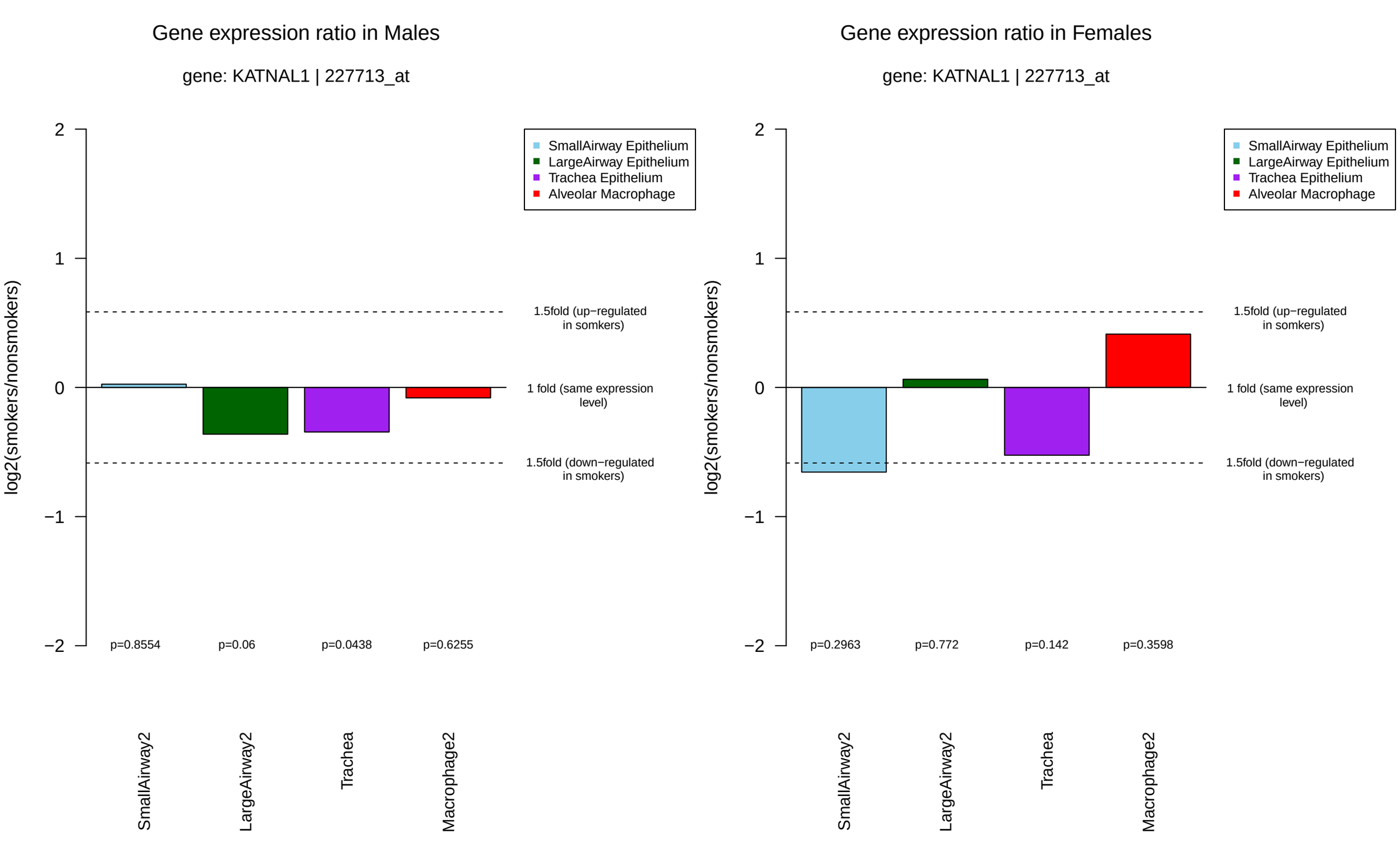
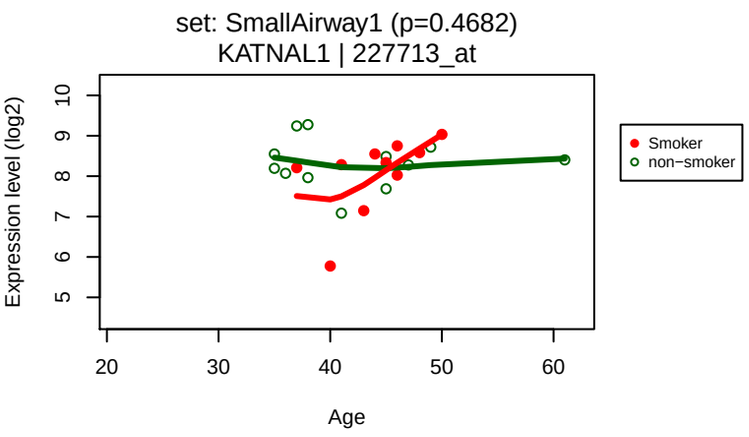
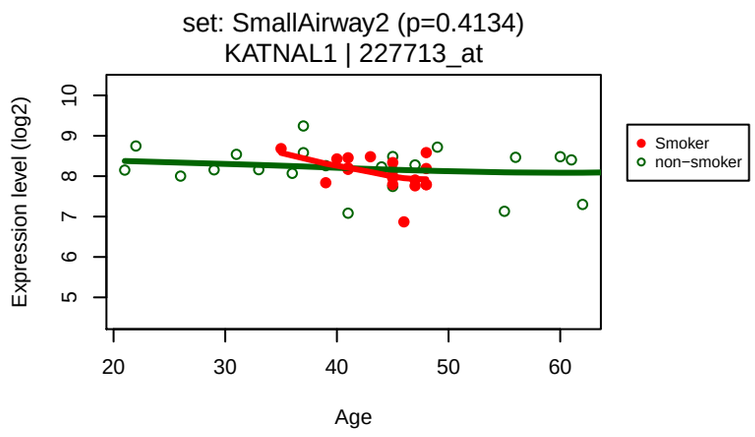
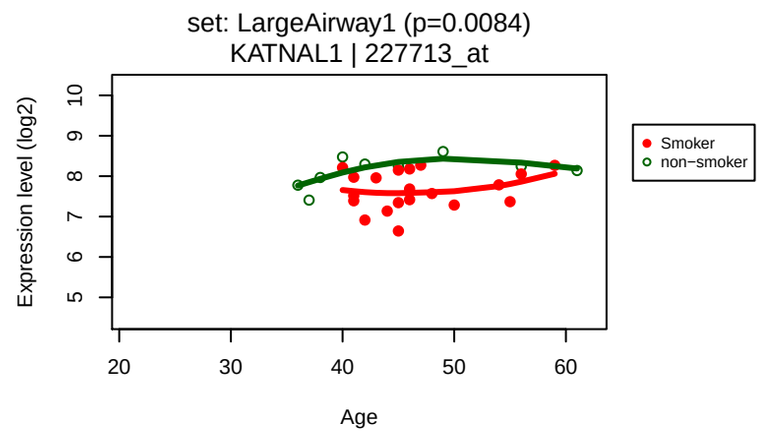

Gene expression of smokers and nonsmokers
gene: KATNAL1 | 227713_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4682
p=0.4134
p=0.034
p=0.0084
p=0.1097
p=0.0087
p=0.7925
p=0.6082
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

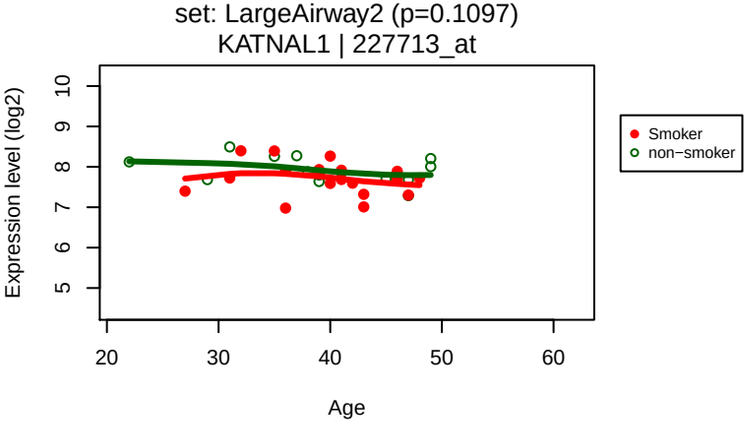

Gene expression ratio (smokers/nonsmokers)
gene: KATNAL1 | 227713_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4682
p=0.4134
p=0.034
p=0.0084
p=0.1097
p=0.0087
p=0.7925
p=0.6082
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

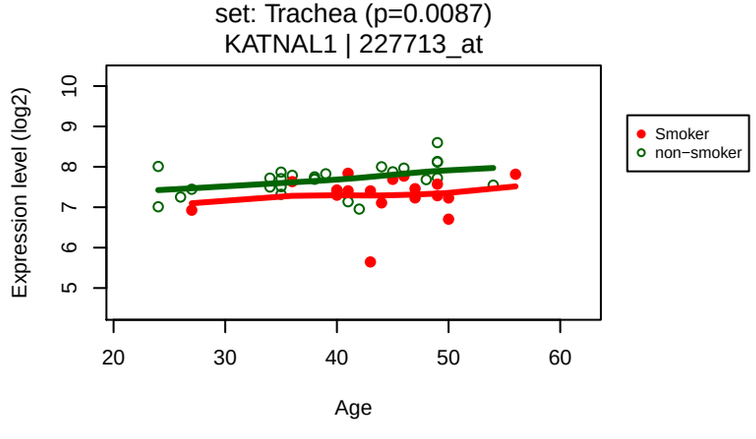

Gene expression ratio in Males
gene: KATNAL1 | 227713_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8554
p=0.06
p=0.0438
p=0.6255
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KATNAL1 | 227713_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2963
p=0.772
p=0.142
p=0.3598
SmallAirway2
LargeAirway2
Trachea
Macrophage2

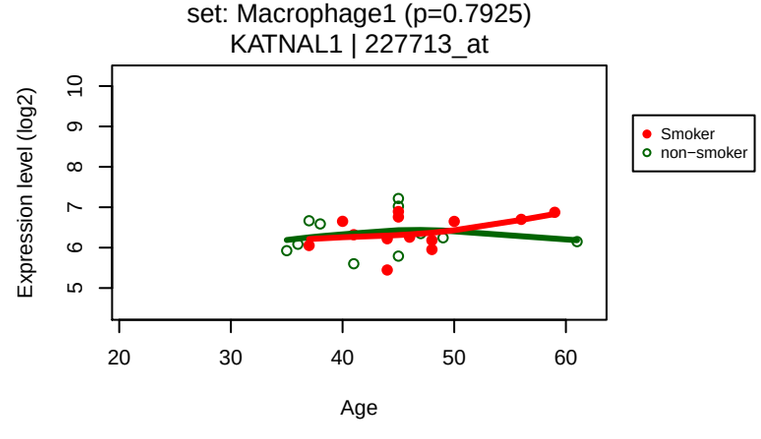

set: SmallAirway1 (p=0.4682)
KATNAL1 | 227713_at
Expression level (log2)
Age
Smoker
non−smoker

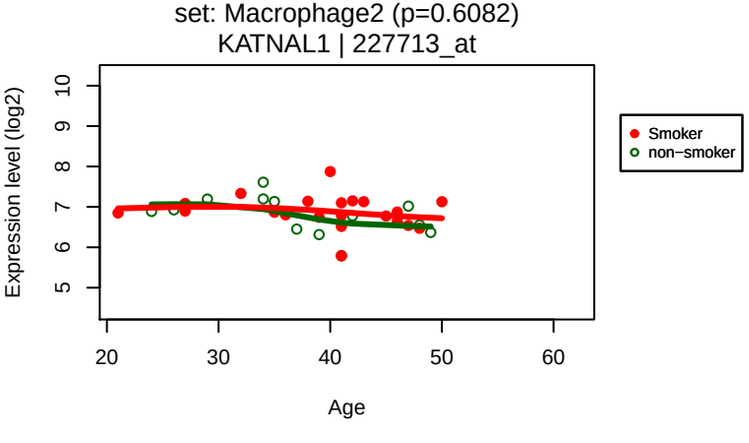

set: SmallAirway2 (p=0.4134)
KATNAL1 | 227713_at
Expression level (log2)
Age
Smoker
non−smoker

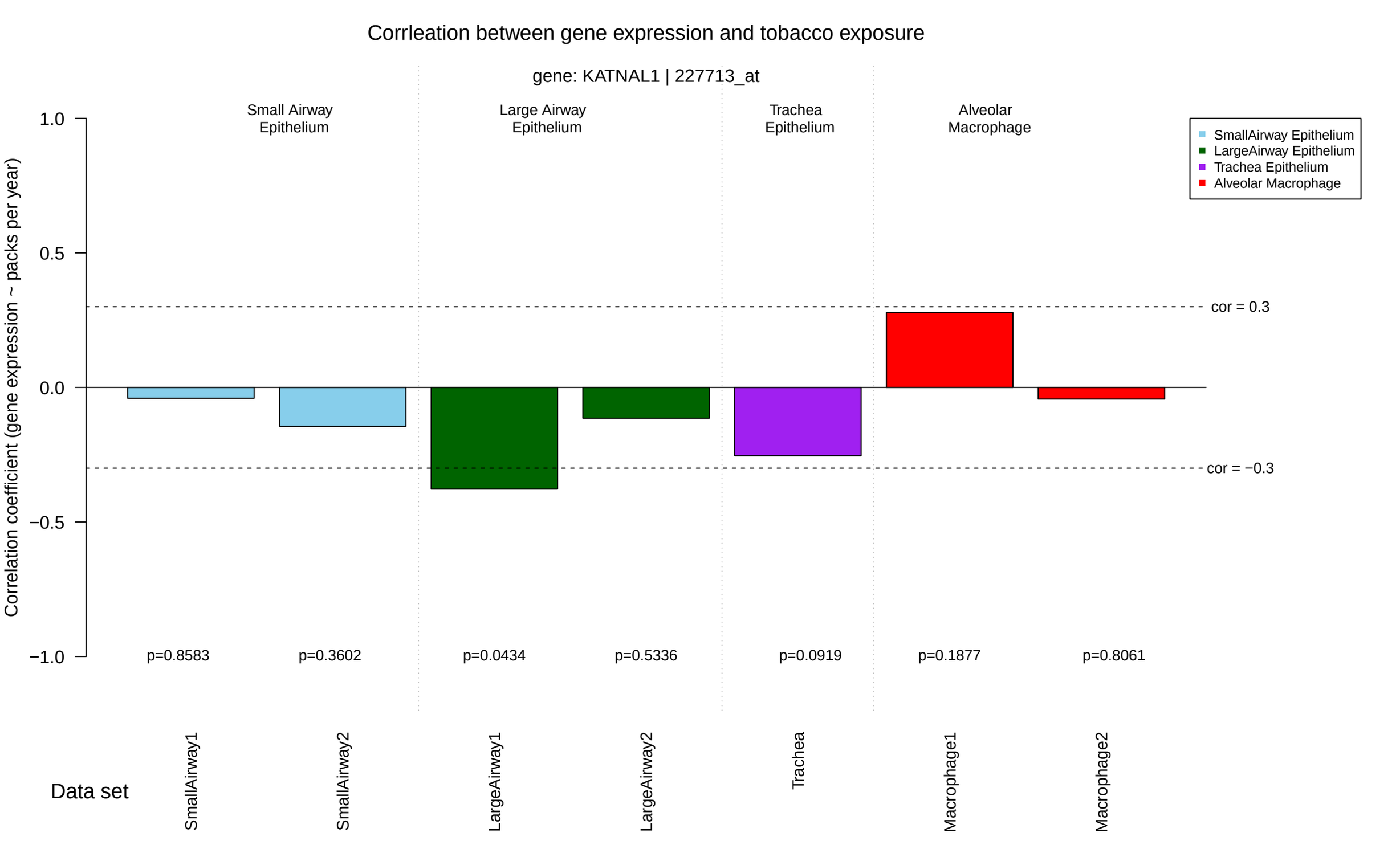

set: LargeAirway1 (p=0.0084)
KATNAL1 | 227713_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1097)
KATNAL1 | 227713_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0087)
KATNAL1 | 227713_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7925)
KATNAL1 | 227713_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6082)
KATNAL1 | 227713_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KATNAL1 | 227713_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8583
p=0.3602
p=0.0434
p=0.5336
p=0.0919
p=0.1877
p=0.8061
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2